

Anuj Chauhan

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PROFESSIONAL SUMMARY

Industrial Simulation Engineer delivering production-validated **physics-based simulation** solutions at Pegatron Corp. I design and deploy **CFD (fluid dynamics)** and **FEA (stress, deformation)** models across the full ANSYS suite — **Fluent, Mechanical, Icepak, LS-DYNA, and Twin Builder** — to solve real thermal, structural, multiphase, and **data center cooling** challenges in high-reliability hardware. Specialized in building **Digital Twins** and **Reduced Order Models (ROMs)** that turn complex 3D simulations into rapid, decision-ready insights. Backed by a **Ph.D. from National Taiwan University**, hands-on semiconductor nanofabrication expertise (EBL, photolithography, RIE/ICP, full cleanroom integration). Bridging the rare gap between **nanoscale fabrication** and **industrial-scale simulation** — ready to apply both to advanced semiconductor manufacturing.

CORE COMPETENCIES & TECH STACK

Semiconductor Fabrication & Metrology

- **Lithography:** Electron Beam Lithography (EBL), Advanced Photolithography, PR Optimization, CD Control.
- **Cleanroom Processing:** End-to-end Process Integration, PVD/CVD, ALD, Plasma Etching (RIE/ICP), Wet Processing.
- **Metrology:** SEM, TEM, Optical Inspection, Defect Analysis, Yield Optimization, Critical Dimension (CD) Validation.

Process Simulation & Modeling

- **Focus:** Multi-Physics Modeling (Thermal/Stress), Data Center Cooling & Thermal Management, Structural Reliability, Physics-Based Digital Twins, DOE, ROM.
- **Stack:** ANSYS (Mechanical, Fluent Multiphase VOF, Icepak, Heatsink, LS-DYNA, Twin Builder, EnSight, ParaView), PyTwin, NVIDIA Omniverse, PyANSYS, SolidWorks, SpaceClaim.
- **Industry Proven:** All simulation capabilities applied and validated on real industrial projects at Pegatron Corp — from concept to production-ready digital twin deployment.

PROFESSIONAL EXPERIENCE

PEGATRON CORPORATION

Section Manager, Software R&D
Technical Manager, Software R&D

Taipei, Taiwan

Mar 2025 – Present

Jun 2024 – Mar 2025

- **Multi-Physics Simulation & Reliability:** Executed advanced ANSYS Mechanical and Fluent simulations for industrial projects, performing high-fidelity stress, deformation, and CFD analysis to optimize structural integrity and fluid flow.
- **LS-DYNA Simulation:** Developed non-linear FEA models in LS-DYNA for large-scale deformation and crash analysis, integrating simulation outputs directly into digital twin workflows.
- **Data Center Thermal & Cooling Simulation:** Developed high-fidelity CFD models using ANSYS Fluent and Icepak to simulate airflow, heat dissipation, and cooling efficiency across data center rack and server environments; delivered actionable insights for thermal management optimization and energy efficiency improvements.
- **Reduced Order Modeling (ROM):** Engineered computationally efficient models using ANSYS Twin Builder to compress complex 3D FEA/CFD results for rapid, accurate reliability assessment.
- **Digital Twin Pipeline:** Spearheaded development of physics-based simulation workflows enabling predictive monitoring of industrial hardware performance under varying structural and thermal constraints.

NATIONAL TAIWAN UNIVERSITY

(Grad. Institute of Photonics & Optoelectronics) — Postdoctoral Fellow

Taipei, Taiwan

Apr 2023 – May 2024

- **Advanced Lithography & Nanofabrication:** Designed and patterned ultra-compact plasmonic waveguides and optical ring-resonators, achieving strict nanoscale tolerances using Electron Beam Lithography (EBL) and precision Photolithography.
- **Process Integration & Optimization:** Directed comprehensive cleanroom process flows (Deposition, Patterning, Etching). Optimized photoresist profiles and plasma etching recipes (RIE/ICP) to significantly improve device yield and reliability.
- **Metrology & Characterization:** Conducted rigorous inspection using SEM and optical metrology tools to validate Critical Dimensions (CD) and ensure high process fidelity across integration cycles.

SHARDA UNIVERSITY

(Department of Physics & Environmental Sciences) — Research Assistant

Greater Noida, India

Jan 2018 – Dec 2018

- Fabrication and characterization of dye-sensitized solar cells and supercapacitors; developed hands-on materials processing and electrochemical characterization skills.

EDUCATION

National Taiwan University

Ph.D., Chemistry

Award: NTU Outstanding International Graduate Student Scholarship

Taipei, Taiwan

2019 – 2023

Gautam Buddha University

M.Sc., Applied Physics

Greater Noida, India

2014 – 2016

University of Delhi

B.Sc., Physics

New Delhi, India

2010 – 2013